4.8 Comparing Linear Functions

Lesson Introduction

Benchmark	MA.912.F.1.5 Compare key features of linear functions each represented algebraically, graphically, in tables, or written descriptions.	Learning Target	• I can compare key features of linear functions.

	Building On	Working Toward
Benchmark Coherence	MA.8.F.1.2 Given a function defined by a graph or an equation, determine whether the function is a linear function. Given an input-output table, determine whether it could represent a linear function.	MA.912.F.1.7 Compare key features of two functions each represented algebraically, graphically, in tables or written descriptions.

Essential Questions	• How can key features of linear functions be compared to one another? • How can key features be compared when the two linear functions are given in different formats? • What components of a linear function determine the end behavior?
Lesson Narrative	Students continue to identify key features, including domain, range, intercepts, and end behavior. End behavior is formally introduced in this lesson, with students making attempts at notation with infinity and its meaning. Key features of linear functions are compared in various forms, including function notation, standard form, table of values, and graphs.

Component Summary	4.8.1 Warm-Up (~5 min.)	Students learn about Dr. Charles Drew and respond to questions interpreting a graph *(SE p. 202)*	4.8.3 Guided Instruction (15 min.)	Formal definition and exploration of "end behavior", including infinity *(SE p. 205)*
	4.8.2 Collaboration (15 min.)	Students explore different representations of the same functions to determine key features *(SE p. 203)*	4.8.4 Your Turn! (10 min.)	Comparing key features of two linear functions *(SE p. 206)*
			4.8.5 Wrap-Up (~5 min.)	Students self-assess their learning progress on each key feature of functions *(SE p. 207)*

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • End behavior <u>Additional Terms:</u> • Intercepts • Slope • Domain • Range	• (optional) Graphing Utility	• Consider preparing small groups or partners based on data from the previous lessons on key features to inform strategic grouping for 4.8.2 Collaboration

Teacher Tips	Common Misconceptions	Things to Consider
	• Students who struggle interpreting the coordinate plane may have difficulty articulating, or they might misinterpret, increasing and decreasing values. • Students may confuse the phrase “largest slope” with “steepest line” without also considering negative and positive slope values. • Students may misinterpret the change in y-values in a given table when the x-values are not consecutive.	• Students use key features like domain and range to make sense of end behavior. Provide scaffolds for students who still struggle with interpreting domain and range to ensure they do not become a barrier to learning end behavior. • This lesson introduces the concept of end behavior. It’s important to stick to the definition provided and within the limits of the linear functions. Students will work with non-linear end behavior in future units.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Think-Pair-Share Question 3 in 4.8.2 Collaboration allows students to engage in a collaboration on comparing representations of linear functions. Consider using the Think-Pair-Share routine to provide students with multiple attempts at making sense of different representations of lines, both independently and with a partner, for increased precision.	**MA.K12.MTR.2.1: Multiple Representations** Clarifications in MA.K12.MTR.2.1 prioritize the use of multiple representations to make sense of abstract concepts, including to “…guide students from concrete to pictorial to abstract representations…” Utilize the discussion prompts in 4.8.2 Collaboration and 4.8.3 Guided Instruction to give students the opportunity to begin transitioning from concrete to abstract.
Differentiate Instruction	This lesson utilizes multiple representations for interpreting key features. Representations include function notation, standard form, written descriptions, tables of values, and graphs. The multiplicity of representations provides a variety of entry points for learners to engage with and make sense of the content. Additional differentiation opportunities are denoted throughout the lesson guide.	
Lesson Notes:		

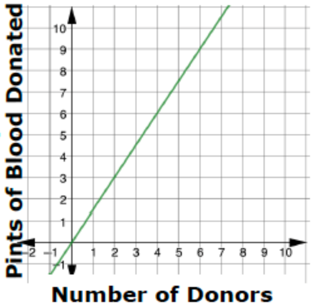
4.8.1 Warm-Up: Blood Donation

Component Overview: *Provide students the opportunity to read the information on Dr. Charles Drew and use discussion questions to build cultural competency before students engage with the graph and questions.*

4.8.1 Warm-Up: Blood Donation

Dr. Charles Drew, a pioneering African American medical researcher, discovered a way to store and process blood for transfusions. Dr. Drew managed two of the largest blood banks in the world during World War II as the medical director of the Red Cross National Blood Donor Service, collecting tens of thousands of pints of blood for U.S. soldiers. It has been said that Dr. Drew's research and work saved the world against Nazism, as battlefield blood storage and transfusions did not exist before. After a ruling stated that the blood from African-Americans could not be transfused to white soldiers, Dr. Drew resigned from the Red Cross and became a professor at Howard University's medical school (an HBCU), training a generation of black physicians on the research of storing and collecting blood.

1. The graph shows the relationship between pints of blood collected and the number of donors according to the Red Cross in the U.S. today.



   a. What is the slope of the line? $\frac{3}{2}$
   b. What does the slope represent? *The slope is* $\frac{3}{2} = 1.5$. *For each donor,* 1.5 *pints of blood are donated.*
   c. Write an equation to represent the linear relationship between the number of donors, d, and the pints of blood collected, b. $b = \frac{3}{2}d$
   d. Use the equation to find the number of pints of blood collected if $12{,}000$ people donate blood.
   $b = \frac{3}{2}d = \frac{3}{2}(12000) = 18{,}000$ *pints of blood collected*

Consider extending this information into a discussion to cultivate cultural competency for your students or partner with an English teacher for more time.

- *Why did Dr. Drew resign?*
- *What does discrimination look, sound, or feel like in today's society?*
- *Dr. Drew's resignation was a form of activism during that time. What does activism look or sound like in today's society?*
- *What does activism look or sound like for you?*

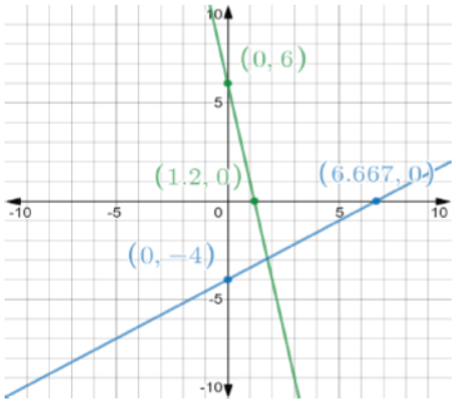
HBCU: Historically Black College or University
At a time when many schools barred their doors to Black Americans, these colleges offered the best, and often the only, opportunity for higher education. There are over 100 HBCUs in the U.S. today still providing an excellent college education to students from diverse backgrounds.

4.8.2 Collaboration: Comparing Representations of Linear Functions

Component Overview: *Students analyze linear functions from different representations to determine and compare key features.*

4.8.2 Collaboration: Comparing Representations of Linear Functions

1. Two linear functions are shown on the coordinate grid.



a. Which line has the larger x-intercept?
Blue

b. Which line has the smaller y-intercept?
Blue

c. Which line has the larger slope?
Blue

d. For which line does y decrease as x increases?
Green

Students should be collaborating throughout the component. Monitor and listen for common student misconceptions to identify which students surface those misconceptions and which students help clarify. Students often confuse 'largest' slope with 'steepest' slope.

Students may pause when determining how to interpret what it means to have a "larger" slope. Consider the following questions for scaffolding:

- *What does it mean to have a "large" slope?*
- *Which slope is larger, 2 or -3? Why?*
- *Based on your answer, which has a larger slope, the green or blue line?*

4.8.2 Collaboration: Comparing Representations of Linear Functions (continued)

2. The table shows other ways the linear relationships could be represented.

	Function	Table of Values	Written Description
Green Line	$f(x) = 1 - 5(x - 1)$	x: -2, 1.2, 2, 7; y: 16, 0, -4, -29	A line with a slope of -5 that passes through the point $(1, 1)$.
Blue Line	$g(x) = \frac{3}{5}x - 4$	x: -5, -1, 0, 3; y: -7, -4.6, -4, -2.2	A line that passes through the points $(0, -4)$ and $\left(6\frac{2}{3}, 0\right)$.

a. With your partner, discuss which representation(s) of the green line can be used to determine the x-intercept.
The table of values

b. With your partner, discuss which representation(s) of the blue line can be used to determine the y-intercept.
The function, the table of values, and/or the written description

3. Brenda and Austin were discussing the representations of the blue and green line.

Brenda	Austin
I can determine whether the linear relationship is increasing or decreasing as x increases using the slope of the line. If the representation does not have the slope, I have to graph the relationship.	*I can determine whether the linear relationship is increasing or decreasing as x increases using the graph, slope-intercept form, point-slope form, or the table.*

Consider opportunities to connect values and key features to their visual representations on the graph in question 1. Particularly for conceptualizing the meaning of phrases like "as x increases" from question 1d above, use an online graphing utility to highlight the coordinate pairs from the table of values for students to connect abstract to concrete.

Which representation made it easier to determine intercepts, the graph in question 1 or the table of values and function notation in question 2? Why?

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.
Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address.

Think-Pair-Share
Provide students with the opportunity to independently read, process, and answer the prompts in question 3 before discussing with a partner. Think-Pair-Share gives students three opportunities to make sense of the information, first independently, then pairing for verbal processing to give an opportunity to build confidence, and finally to generate precise responses when students share out.

4.8.2 Collaboration: Comparing Representations of Linear Functions (continued)

a. With your partner, discuss how Brenda can determine if a linear relationship is increasing or decreasing using only the slope of the line. Write a summary of your discussion.
Answers will vary. A positive slope means the linear relationship is increasing because the y-values increase as the x-values increase. A negative slope means the linear relationship is decreasing because the y-values decrease as the x-values increase.

b. With your partner, discuss how Austin could use a table to determine if a linear relationship is increasing or decreasing. Write a summary of your discussion.
Answers will vary. Order the x-values from smallest to largest and use the corresponding y-values to determine if the relationship is increasing or decreasing.

c. Brenda told her teacher that she could not determine if the blue line is increasing or decreasing from its written description because it did not include the slope. Explain to Brenda how she could use the written description to find the slope.
Brenda can use the two points to find the slope of the line.

d. Work with your partner to help Brenda and Austin understand how to use an equation in standard form, such as the equation that represents the blue line, $3x - 5y = 20$, to determine if the relationship is increasing or decreasing.
Rewrite the equation $3x - 5y = 20$ in slope-intercept form.
$-5y = -3x + 20$
$y = \frac{3}{5}x - 4$
Since the slope is positive, then the y-values increase as the x-values increase.

Students may believe that a "larger" slope means that the linear relationship is increasing. Ensure that students recognize that the lines with a negative slope will always be decreasing from left to right and lines with a positive slope will always be increasing from left to right.

Discussion provides additional entry points for auditory learners. Consider utilizing sentence stems and providing time for students to grapple with academic language:

- *I think the linear relationship is _______ because the slope is _____*
- *I notice in the table that _______ which makes me think _____*

For question 3a, would a horizontal line be increasing or decreasing? (a horizontal line has a slope of 0 so it is neither increasing nor deceasing).
For question 3d, consider asking students whether slope can be determined from standard form.

4.8.3 Guided Instruction: Comparing End Behavior

Component Overview: *Students will learn how to compare two different linear functions represented in different ways. They will also be formally introduced to the concept of end behavior.*

4.8.3 Guided Instruction: Comparing End Behavior

Two different linear functions are shown.

Line A		Line B
x	y	$f(x) = -\frac{4}{9}x + 2$
-18	4	
-12	$\frac{4}{3}$	
0	-4	
2.25	-5	
18	-12	

1. Which linear relationship has the smaller y-intercept?
   Line A

2. Compare the domains of the two linear relationships.
   The domains are the same.

3. Determine which linear relationship has the larger slope.
   The slopes are the same.

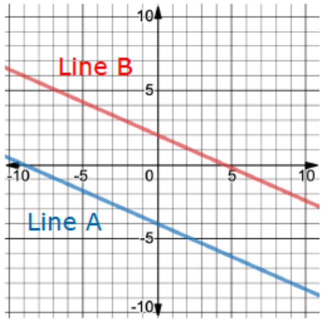
4. Determine if line A is increasing or decreasing as x increases. Show your work or explain your thinking.
   The slope is negative, so Line A is decreasing as x increases.

5. Determine if line B is increasing or decreasing as x increases. Show your work or explain your thinking.
   The slope of Line B is negative, so Line B is decreasing as x increases.

Begin by introducing the two linear functions that will be used and guide students through how to compare two different linear functions. Because it is stated that these are linear functions students can assume that the pattern in the table will continue. Students are taught in 8th grade that that a table may not determine a function (MA.8.F.1.2) so the statement before the two lines, "Two different linear functions are shown.", clearly establishes that the table for Line A is a function.

Which representation is easier for determining whether values are increasing or decreasing? Why?

6. Sketch and label the graph of each linear relationship on the coordinate grid below.



What do you notice about the graphs you have sketched? What key features would be the same for the lines?

The ***end behavior*** of a function refers to the appearance of the function's graph at the "ends," in other words, as the x-values get infinitely smaller or as the values get infinitely larger.

End behavior is formally introduced. Students will continue to make sense of end behavior in a variety of contexts, so mastery is not yet expected in this lesson.

7. Use the definition of end behavior to complete the sentences.
   a. For line A, as the x-values get smaller, the appearance of the graph of A approaches
      - ○ $-\infty$.
      - ○ ∞.
   b. For line A, as the x-values get larger, the appearance of the graph of A approaches
      - ○ $-\infty$.
      - ○ ∞.

Introduce the concept of end behavior to students and model question 7 with the additional explanation.

Another way to represent end behavior is to use mathematical notation. Two examples are shown in the table.

Notation	Meaning
$x \to -\infty$	"x goes to negative infinity"
$y \to \infty$	"y goes to infinity"

Think back to what you've learned about the concept of infinity. What does it mean for x to go to negative infinity? What does it mean for x to go to positive infinity?

8. Complete the mathematical notation for end behavior for line B.
   As $x \to -\infty$, $y \to \infty$. As $x \to \infty$, $y \to -\infty$.

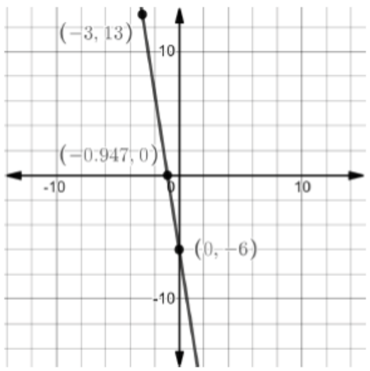
For a challenge, have students determine if they can write a function, using a written description only, that results in the end behavior in question 8.

4.8.4: Your Turn!

Component Overview: *Students work independently to practice comparing key features of two linear functions.*

4.8.4 Your Turn!

1. Two different linear functions are shown.

Line A	Line B
	$b(x) = -7x + 3$

a. Name the key feature(s) that are the same for both linear functions.
End Behavior, Domain, Range

b. Give the key feature(s) that are different for both functions.
Slope, x-intercept, y-intercept

c. Which linear function has the smaller slope?
Line B

d. Which linear function has the larger x-intercept?
Line B

Monitor which representations students are using when on their own to determine the key features to answer the comparison questions. Their approach might illuminate potential gaps or learning styles.

For students who finish early, challenge them to create a third linear function that shares only two key features with Line A. Consider making the challenge harder by specifying which elements need to be the same and which should be different.

4.8.5 Wrap-Up: Reflection

Component Overview: *Students complete a self-assessment based on their level of understanding of each key feature. It is designed to be done independently to gather specific student-level data.*

4.8.5 Wrap-Up: Reflection

1. Complete the table to describe your understanding of each key feature.
   Answers will vary.

Domain	Range
☐ I need help. ☐ I got it. ☐ I could help someone.	☐ I need help. ☐ I got it. ☐ I could help someone.
Slope	**x-intercept**
☐ I need help. ☐ I got it. ☐ I could help someone.	☐ I need help. ☐ I got it. ☐ I could help someone.
End Behavior	**y-intercept**
☐ I need help. ☐ I got it. ☐ I could help someone.	☐ I need help. ☐ I got it. ☐ I could help someone.

Consider using student responses from prior self-reflections to consider a student's learning progress. Recording this information and pairing it with assessment data provides insightful feedback for student learning progression.